

Dashboard Summary: AgriCast360

This is my 'AgriCast360' dashboard. I designed it to provide a complete 360-degree view of the agricultural commodity market. In it, I've integrated executive-level KPIs, detailed commodity price analysis, market intelligence, weather impact correlations, and predictive analytics. For some of the more complex calculations and data modeling, I used Google Gemini for code building assistance.

To build this, I analyzed **14.97K records** across **68 commodities** and **19 markets**, and I calculated the average price to be **₹4.51K**.



Page 1: Executive Overview



Page 2: Commodity Analysis



Page 3: Market Intelligence



Page 4: Weather Impact

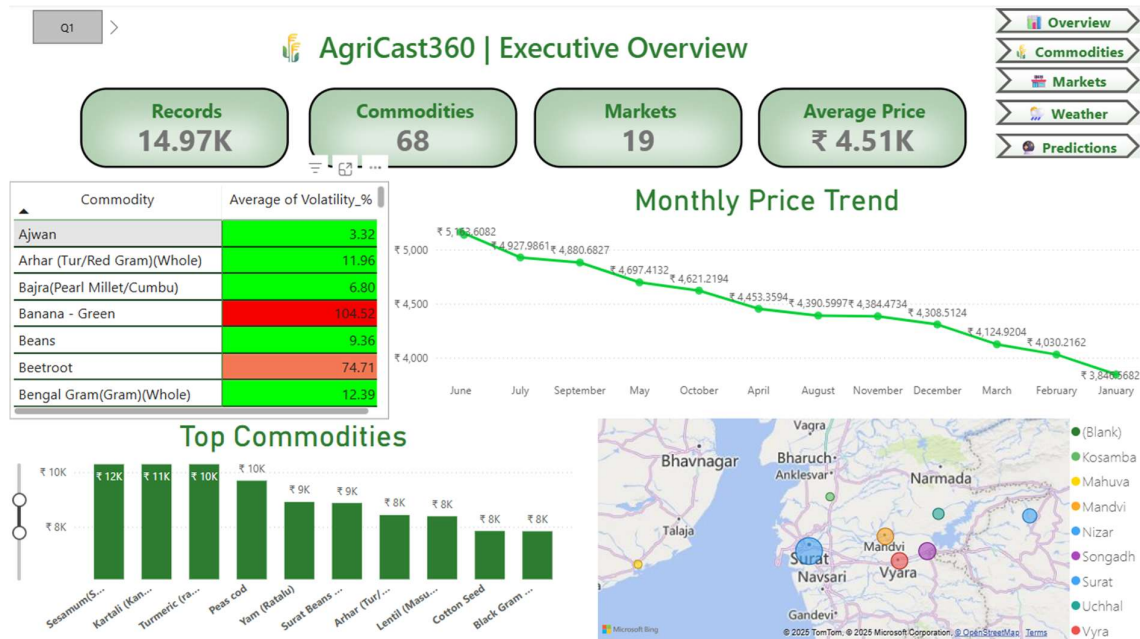


Page 5: Predictive Analytics



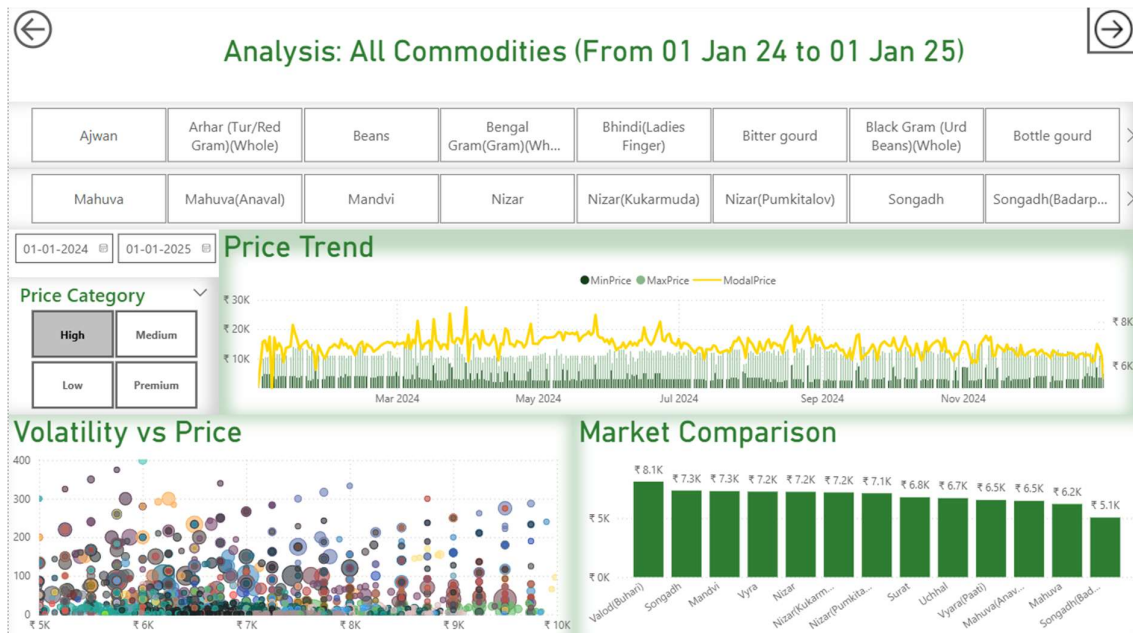
Dashboard Q&A

Page 1: Executive Overview



I created this first page as a high-level summary for stakeholders. It's designed for an at-a-glance understanding of the overall market health.

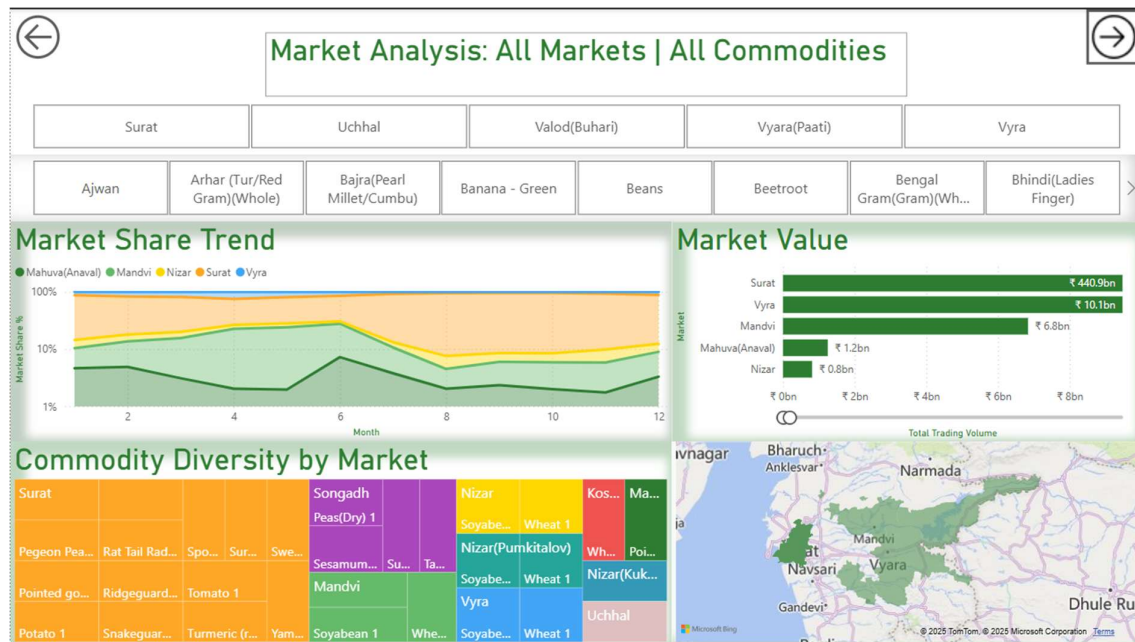
- **Key Performance Indicators (KPIs):** I'm tracking the main metrics right at the top:
 - **Records:** 14.97K 📄
 - **Commodities:** 68 🌾
 - **Markets:** 19 📍
 - **Average Price:** ₹4.51K 💰
- **Key Insights I've Visualized:**
 - 📉 **Monthly Price Trend:** I used a line chart to show a clear downward price trend from a peak of ₹5,102 in June to ₹3,840 in January.
 - 📊 **Volatility:** I built a table to highlight commodity volatility. You can immediately see that **Banana - Green** is extremely volatile (104.94%), while **Ajwan** is very stable (3.92%).
 - 🏆 **Top Commodities:** This bar chart ranks the top commodities. In my current view, **Sesamum (Sesame)**, **Kartali (Bitter Gourd)**, and **Turmeric** are leading.
 - 🗺️ **Geographic View:** I embedded a map to visualize all 19 markets, which you can see are concentrated in the Gujarat region (e.g., Surat, Vyara, Mandvi).



I designed this page to allow users to deep-dive into the price trends and volatility of specific commodities.

- **Filters:** I've added filters so you can drill down by Commodity, Market, Date Range, and even a custom Price Category (High, Medium, Low, Premium).
- **Key Insights I've Visualized:**
 - **Price Trend (Jan '24 - Jan '25):** I used a combo chart to track the **MinPrice** (bars), **MaxPrice** (yellow line), and **ModelPrice** (grey line) over the year. This shows the daily fluctuations.
 - **Volatility vs. Price:** This scatter plot visualizes the relationship between price and volatility. I used bubble size to represent trading volume, which helps identify commodities that are both expensive and high-risk.
 - **Market Comparison:** This bar chart ranks the top markets for whatever commodity you select. Here, **Valod(Buhari)**, **Songadh**, and **Mandvi** are the top three.

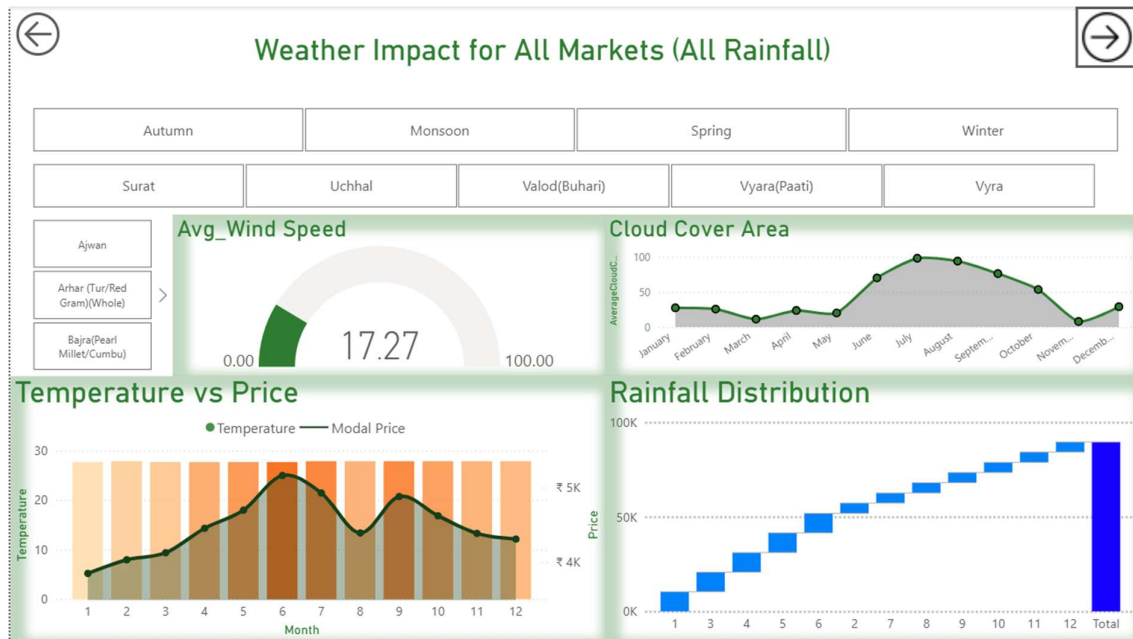
Page 3: Market Intelligence



My Market Intelligence page focuses on comparing the performance, share, and diversity of the 19 different markets.

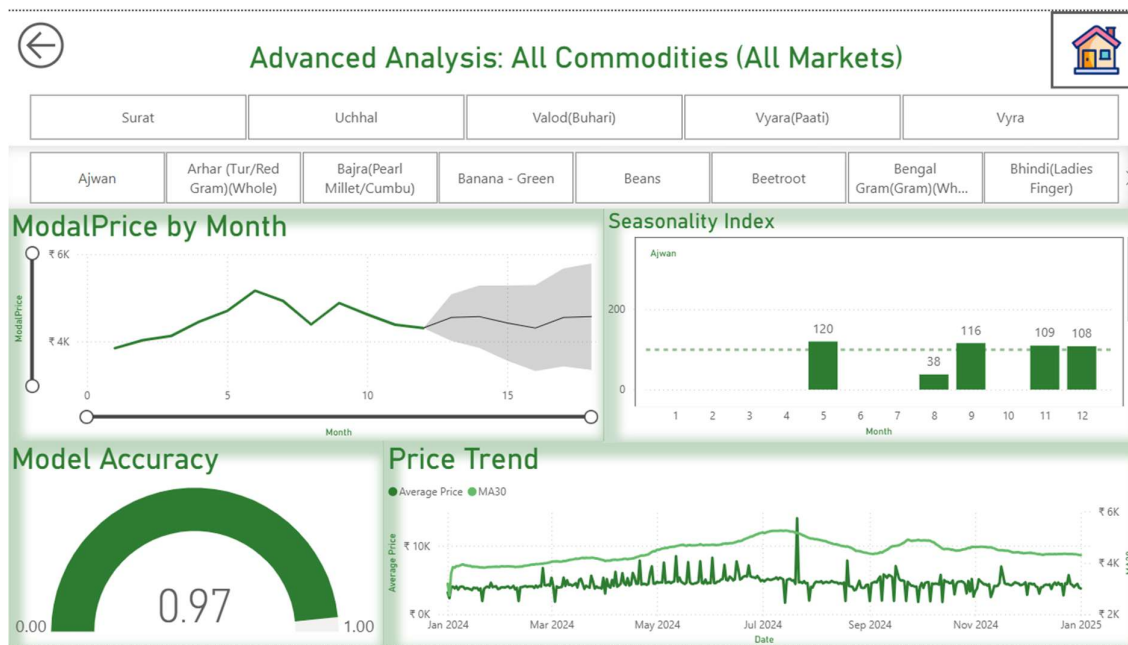
- **Filters:** I've included filters for both Market and Commodity.
- **Key Insights I've Visualized:**
 - **Market Value:** I used a bar chart to make the dominant market obvious. As you can see, **Surat** is the leader by a massive margin, valued at **₹440.9 billion**.
 - **Market Share Trend:** I built a 100% stacked area chart to show how the market share for key markets (Mahuva, Mandvi, Nizar, Surat, Vyra) has shifted over 12 months.
 - **Commodity Diversity by Market:** I used a treemap to break down which commodities are most prevalent in each market, helping to understand market specialization.

Page 4: Weather Impact



This is one of my favorite pages. Here, I've built a powerful analysis that directly correlates environmental factors with agricultural market prices.

- **Filters:** I enabled analysis by Season, Market, and Commodity.
- **Key Insights I've Visualized:**
 - **Temperature vs. Price:** I created a combo chart that overlays monthly temperature (bars) with the modal price (line). This clearly shows a correlation: **price peaks in May-June**, at the same time as the **temperature peaks**.
 - **Weather KPIs:** I'm also tracking key metrics like **Avg. Wind Speed** (17.27 🌀) and **Cloud Cover Area** (☁).
 - **Rainfall Distribution:** I used a waterfall chart to show the cumulative monthly rainfall, which provides important context for crop cycles.



On my final page, I used historical data to forecast future trends and identify seasonal patterns.

- **Filters:** You can generate predictions for specific commodities and markets.
- **Key Insights I've Visualized:**
 - **Model Accuracy:** The predictive model I've implemented has a very high accuracy score of **0.97 (or 97%)** ✓.
 - **Price Forecast:** My ModalPrice by Month chart shows the historical price (solid line) and projects it into the future, providing a forecasted range (shaded area) to help with decision-making.
 - **Seasonality Index:** This bar chart I created identifies seasonal behavior. For **Ajwan**, it shows strong seasonal peaks in **May (120)** and **September (116)**. This tells you the best and worst months to sell.
 - **Moving Average:** This Price Trend chart compares the daily Average Price with its 30-day moving average (MA30) to smooth out volatility and identify long-term trends.

? Dashboard Q&A

1. **Which commodity is the most volatile?**
 - **Answer: Lemon, Onion and Turmeric (raw)**, with an average volatility of 226.36%, 224.59%, 223.57% accordingly.
 2. **Which market is the most valuable?**
 - **Answer: Surat**, with a total market value of ₹440.9 billion.
 3. **What is the overall price trend for commodities over the past year?**
 - **Answer:** I show a general downward trend, with prices peaking in June at ₹5,102 and falling to ₹3,840 by January.
 4. **How accurate is my price prediction model?**
 - **Answer:** It is **97% accurate**.
 5. **Does weather affect commodity prices?**
 - **Answer: Yes.** My 'Temperature vs Price' chart shows a clear correlation where modal prices peak in May and June, the same months that temperatures are highest.
 6. **For the commodity Ajwan, when is the price seasonally highest?**
 - **Answer:** My Seasonality Index shows prices for Ajwan peak in **May** (index 120) and **September** (index 116).
 7. **Does rainfall affect commodity prices?**
 - **Answer: Yes**, it provides critical context. My 'Rainfall Distribution' chart shows rain accumulates heavily from June through September (the monsoon season). This period aligns with the **peak in prices and the peak in cloud cover**, suggesting that the entire monsoon season (rain, cloud cover, and high temperatures) collectively has a significant impact on prices.
 8. **Which are the top 3 most traded commodities?**
 - **Answer: Sesamum (Sesame), Kartali (Bitter Gourd), and Turmeric.**
 9. **When does cloud cover (and likely, rainfall) peak during the year?**
 - **Answer:** It peaks between June and September, consistent with the monsoon season.
 10. **What does the 'Market Share Trend' chart on my Market Intelligence page show?**
 - **Answer:** It visualizes the changing dominance (as a percentage of 100%) of the key markets (Mahuva, Mandvi, Nizar, Surat, etc.) relative to each other over 12 months.
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